

Worksheet / Werkstuk 4.4 - 4.5

A

Evaluate / Bepaal

$$1 \quad \lim_{x \rightarrow \pi} \frac{\tan x}{x}$$

$$6 \quad \lim_{x \rightarrow \pi/2} (\sec x - \tan x)$$

$$11 \quad \lim_{x \rightarrow \infty} \frac{\ln(\ln x)}{\ln x}$$

$$2 \quad \lim_{x \rightarrow 0} \frac{x - \sin x}{x - \tan x}$$

$$7 \quad \lim_{x \rightarrow \infty} (e^{2x} + 3x)^x$$

$$12 \quad \lim_{x \rightarrow \pi/2^-} (\sin x)^{\tan^2 x}$$

$$3 \quad \lim_{x \rightarrow \infty} \frac{3 - e^{3x^2}}{3x - 6}$$

$$8 \quad \lim_{x \rightarrow \infty} x^3 e^{-x}$$

$$13 \quad \lim_{x \rightarrow 0^+} \left(\frac{1}{x}\right)^{\sin x}$$

$$4 \quad \lim_{x \rightarrow -\infty} \frac{e^{3x^2}}{3x}$$

$$9 \quad \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$$

$$14 \quad \lim_{x \rightarrow \infty} (e^{2x} + 3x)^{\frac{1}{x}}$$

$$5 \quad \lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}}$$

$$10 \quad \lim_{x \rightarrow \infty} x \ln\left(1 + \frac{1}{x}\right)$$

$$15 \quad \lim_{x \rightarrow -\infty} \frac{\sqrt{7 + 9x^2}}{1 - 2x}$$

B

1. Sketch the graph of $f(x) = \frac{6}{4 - 2^x}$ using the guidelines for sketching a curve

Skets die grafiek van $f(x) = \frac{6}{4 - 2^x}$ deur gebruik te maak van die riglyne vir die skets van krommes

2. Determine the local extrema of the function $f(x) = x^3 - x^2 + 2$.

Bepaal die lokale ekstreme van die funksie $f(x) = x^3 - x^2 + 2$.

3. Determine the concavity and inflection points of $f(x) = x^4$.

Bepaal die konkawiteit en infleksie punte van $f(x) = x^4$.